

AI GOVERNANCE PIPELINE

16-Week Development Roadmap

From environment setup to a rehearsed live demo — eight phases, four detectors, one explainable risk score.



3-person team, roughly full-time · stretch to ~20 weeks if part-time

The 16-Week Timeline

Eight phases, each with a hard exit criterion before the next begins.



Phases 1–4 · Building the Pipeline

Weeks 1–10 · foundations through automated risk decisions

1 Foundations WEEKS 1–2

- ✦ W1 — Env & repo setup: FastAPI scaffold, Git branching, .env config, PostgreSQL, SQLAlchemy models drafted
- ✦ W2 — Local LLM baseline: Ollama installed, qwen3:8b standalone

EXIT /health endpoint live; Qwen answers a raw prompt via Ollama outside the app.

2 Core Security Detectors WEEKS 3–5

- ✦ W3 — Prompt Filter Layer: keyword/rule blocklist, category tagging, unit tests
- ✦ W4 — Prompt Injection Detector: regex + pattern library vs. HF benchmark dataset
- ✦ W5 — PII Detector: custom recognizers, precision check

EXIT All three detectors independently unit-tested, returning structured, explainable output.

3 Guardrails & LLM Integration WEEKS 6–8

- ✦ W6 — NeMo Guardrails config: config.yml, Ollama as engine, integration test
- ✦ W7 — Pipeline wiring: steps 1–7 chained in main.py, graceful fallback if unreachable
- ✦ W8 — Latency & reliability pass: timeouts, retries, p95 benchmarking

EXIT A full prompt travels filter → injection → PII → guardrails → Qwen and back with a real response.

4 Risk & Decision Engine WEEKS 9–10

- ✦ W9 — Risk scoring: weighted model (Injection 70 / Filter 40 / NeMo 30 / PII 20), reason aggregation
- ✦ W10 — Decision thresholds: Allow 0–29 / Review 30–69 / Block 70–100, calibration

EXIT Given any prompt, the API returns a score, a level, a decision, and why.

Phases 5–8 · Proving It Works

Weeks 11–16 · audit trail through the final live demo

5 Audit Logging & Persistence

WEEKS 11–12

- ✦ W11 — Database schema: Alembic migrations, audit table, SQLAlchemy models finalized
- ✦ W12 — Audit + reporting endpoints: write path into /validate, read/query routes, compliance report

EXIT Every /validate call is durably logged and retrievable — a real audit trail.

6 Evaluation & Benchmarking

WEEKS 13–14

- ✦ W13 — Batch evaluation run: full pass of the ~5,000-prompt benchmark, results logged
- ✦ W14 — Metrics & tuning: precision/recall/F1 per detector, false-positive review, re-tuning

EXIT A defensible accuracy figure — caught X% with Y% false positives.

7 Integration Testing & Hardening

WEEK 15

- ✦ W15 — End-to-end + security pass: full pytest suite, OpenAPI docs cleanup, Cyshield security review, edge-case testing

EXIT No known crashes; Cyshield sign-off on the security posture.

8 Final Packaging & Demo

WEEK 16

- ✦ W16 — Deployment prep + defense materials: Docker packaging (or documented local-run), final demo script, report + slide polish

EXIT A working, presentable system plus a rehearsed live demo.

- The weekly meeting of Queen's University on Tuesday at 8:00 am – Toronto time
- The weekly meeting of Cyshield Company on Wednesday at 3:00 pm – Cairo time

Project Supervision



This Project is monitored by Eng. Abraam Nagieb , Eng.Samy Ezzat and Research and Development team

Dr.Ahmed Saafan , Dr.Heba Osama and Sara Ali (Cyshield Company)



This Project is supervised by Prof . Mohammad Zulkernine (Queen's University)



Our team work

Bassant Yehia Madkour , Mariam Tanious , Mohamed Salah